

Cybersecurity Foundations



Securing a Computer
System



Securing a Computer System



وزارة الاتصالات
وتكنولوجيا المعلومات



Congratulations!

You have been hired to audit the security for the PC used at your friend's business: Joe's Auto Body. Joe provides car repair services throughout the tri-state area. He's had previous employees use it for activities unrelated to work (e.g., web browsing, personal email, social media, games, etc.), and he now uses it to store his critical business information. He suspects that others may have broken into it and could be using it to transfer files across the internet. He has asked that you secure it for him according to industry best practices so that it can again be used as a standard PC.

In this project, you have been given a "broken" Windows 10 PC and asked to figure out what's wrong with it and then make changes to fix and secure it. The process of analyzing and applying security happens in workplaces around the globe and is exactly what cybersecurity professionals do daily. This project allows you to apply what you've learned in the course by investigating a Windows 10 PC. The same skills you use on one PC can be applied to thousands.

You do not need to do anything on this slide.



Part 1: Reconnaissance



Hardware

The first step in securing any system is to know what it is, what's on it, what it's used for, and who uses it. That's the concept of systems reconnaissance and asset inventory. In this step, you'll document the hardware, software, user access, system and security services on the PC. Complete each section below.

1	<i>Device Name</i>	JoesGaragePC
2	<i>Processor</i>	Intel(R) Xeon(R) CPU E5-2673 v4 @ 2.30GHz
3	<i>Install RAM</i>	4.00 GB
4	<i>System Type</i>	64-bit operating system, x64-based processor
5	<i>Windows Edition</i>	Windows 10 Pro
6	<i>Version</i>	22H2
7	<i>Installed on</i>	11/23/2021
8	<i>OS build</i>	19045.2486



Software

Another common early step in securing is taking an inventory of software or applications installed on a computer system. These are programs outside of the standard operating system. Please list five applications running on this PC.

1	Adobe Reader XI (11.0.01)
2	Candy Crush Friends
3	Google Chrome
4	7-Zip 19.00 (x64)
5	Microsoft 365 Copilot



Security Services

Document the PC's security settings status listed below.

<i>Security Feature</i>	<i>Status</i>
<i>Firewall product and status--Private network</i>	Product: Microsoft Defender Firewall Private Network Status: OFF (Inactive and Vulnerable)
<i>Firewall product and status--Public network</i>	Product: Microsoft Defender Firewall Public Network Status: ON (Active and Protected)
<i>Virus protection product and status</i>	Product: Microsoft Defender Antivirus Status: Active and Up to Date
<i>Internet Security messages</i>	Warning: Reputation-based protection is(off.) Device may be vulnerable



<i>Security Feature</i>	<i>Status</i>
<i>Network firewall messages</i>	Warning: Microsoft Defender Firewall settings may make the device unsafe. Domain network firewall is OFF.
<i>Virus protection messages</i>	Set up OneDrive for file recovery options in case of a ransomware attack. Antivirus is active, but ransomware protection is not set up.
<i>User Account Control Setting</i>	Never notify



Accounts

As part of your security assessment, you should know the user accounts that may access the PC. Please list the accounts, name, and access level for the accounts on this PC.

<i>Account Name</i>	<i>Full Name</i>	<i>Access Level</i>
AUser	A User	Standard User (Cyber Course Account)
DefaultAccount	Not shown	Managed User Account
Frank	Frank	Standard User
Guest	Not shown	Limited Guest Access
Hacker	A Hacker	Standard User
JaneS	Jane smith	IT Manager / Administrator
JoesAuto	Joes Account	Built-in Account for Admin tasks
Notedmin	Do Not Use	Disabled / Restricted Access
WDAGUtiliy	Not shown	Built- in System Account
Administrator		Full System Administrator
ServiceAccount		Used for System Services





Authentication

Consider the 3 factors of authentication: something you KNOW, something you HAVE, and something you ARE. In 1 to 3 sentences below, suggest and explain what type of authentication would be appropriate for JoesPC.

- 1- Something you **KNOW** is a password or PIN.
- 2- Something you **HAVE** is a physical security key or a paired Bluetooth device.
- 3- Something you **ARE** includes biometric authentication like fingerprint or facial recognition.

For Joe's PC, a **multi-factor authentication (MFA) approach** would be appropriate. Using a **password (something you KNOW) along with a security key (something you HAVE)** enhances security. If biometric authentication (something you ARE) is available, it can further improve protection by ensuring only Joe can access the device.



System and Security: Firewall

Please answer the following question.

1 *In 1 to 2 sentences, explain what protection would enabling the Windows Firewall provide.*

1- Enabling Windows Firewall helps protect the system by blocking unauthorized access and potentially harmful network traffic.

2-It acts as a security barrier that prevents hackers, malware, and other threats from compromising the device.



System and Security: Firewall

Scenario: You need to ensure the Windows Defender anti-virus is enabled to always protect against current threats. It should be set to continually scan the PC for malicious software automatically. Please answer the following question.

1 *In 1 to 2 sentences, explain what protection enabling the Microsoft Defender Antivirus would provide.*

1-Enabling Microsoft Defender Antivirus provides real-time protection against viruses, malware, ransomware, and other security threats.

2-It continuously scans files, applications, and downloads to detect and remove harmful software before it can damage the system.



Part 3: Securing Access



Users - Part 1

Ensuring only specific people have access to a computer is a common step in information security. It starts by understanding who should have access and the rules or policies that should be followed. Please review the following users who should have access.

- JoesAuto
- Jane Smith (Joe's Assistant)

It is your responsibility to create suggestions for securing this computer. Use the next slide to give and explain your recommendations. The slide following your recommendations will have two questions regarding users and privileges.



Users - Part 2

Fill in this table based on the guidelines you would recommend to Joe. Recommendations do not have to be in complete sentences. Explanations must be at least one sentence.

	<i>Recommendation</i>	<i>Explanation</i>
<i>How should users authenticate their identity?</i>	Use strong passwords and enable multi-factor authentication (MFA).	Strong passwords help prevent unauthorized access, and MFA adds an extra layer of security
<i>What Access Rights/Permissions should Joe have?</i>	Administrator access.	As the manager, Joe needs full control over the system, including user management, software installation, and security settings. His Administrator role allows him to perform these tasks efficiently.
<i>What Access Rights/Permissions should Jane have?</i>	Standard user access with necessary permissions for her tasks.	Since Jane is an assistant, she should not have full administrator access. Instead, she should be granted permissions to perform her job



Users- Part 3

Please answer the following two questions.

1 *In 1 to 2 sentences, explain why it is important to disable or remove unneeded accounts from a PC or application.*

Reduces Insider Threats: Former employees or users with lingering accounts may still have access to sensitive data, posing a security risk.

Minimizes System Vulnerabilities: Fewer active accounts mean fewer potential entry points for cyber threats, improving overall security.

2 *Administrator privileges for too many users is a security challenge. Provide at least 3 risks associated with users having administrator rights on a PC.*

1.Unauthorized User Access (Security Breach Risk)

2.System Configuration Changes (Integrity & Stability Risk)

3.Remote Access Vulnerabilities (Unauthorized Control Risk)



Part 4: Securing Applications



Unnecessary Applications

Joe wants everyone to use the latest version of the Internet Explorer browser by default. There should be no games or non-work-related applications installed or downloaded. Joe is also concerned that there are “hacking” programs downloaded or installed on the PC that should be removed. This PC is used for standard office functions.

1 *List three applications that violate this policy.*

1. **Candy Crush Friends** – This is a game and not related to work.
2. **Farm Heroes Saga** – Another game that is non-work-related.
3. **Nmap 7.80** – This is a network scanning tool often used for hacking or penetration testing, which violates Joe’s concern about hacking programs.

2 *Name three vulnerabilities, threats, or risks to having unnecessary applications.*

1. Security Risks (Nmap, Npcap, VNC Server & Viewer)
2. Productivity Loss (Candy Crush Friends, Farm Heroes Saga, Spotify)
3. System Performance Issues (Multiple Unnecessary Applications)



Patching and Updates

All applications should be up-to-date on patches or fixes by the manufacturer. Any old version of software should be uninstalled. List two applications on JoesPC that are out of date.

1	Adobe Reader XI (11.0.01) – Last updated on 5/11/2020
	This version is outdated and may contain security vulnerabilities that could be exploited.
2	Google Chrome – Last updated on 5/11/2020
	An outdated browser poses security risks, including exposure to malware, phishing, and compatibility issues with modern web applications.



Standout Suggestions



Standout Suggestion 1

Joe has decided to allow least privilege access to 2 additional employees. He would like the bookkeeper and the head mechanic to have access to JoesPC. In 3 - 5 sentences total below, describe the privileges these two employees should have, and detail how they should authenticate their identities.

- Bookkeeper:** Should have access to financial records, accounting software, and any bookkeeping-related applications. They should not have administrative rights to install or remove programs. Authentication should be done using a strong password and, if possible, multi-factor authentication (MFA) for added security.

Head Mechanic: Should have access to any software related to vehicle diagnostics, inventory tracking, and maintenance schedules. They should not have access to financial records or administrative system settings. Authentication should also be through a unique password and, if available, MFA.



Standout Suggestion 2

Joe believes one of his employee's emails has been compromised. What are the possible threats, risks, or vulnerabilities, and how should he respond? Detail your answer in 3 to 5 sentences.

- 2 **Threats and Risks:** A compromised email can lead to phishing attacks, data breaches, or unauthorized access to sensitive company information. Attackers might use the email to send fraudulent messages, distribute malware, or gain access to other systems through password resets.
- Immediate Response:** Joe should instruct the employee to change their email password immediately and enable multi-factor authentication (MFA). IT should scan the network for any suspicious activity and check email logs for unauthorized access.